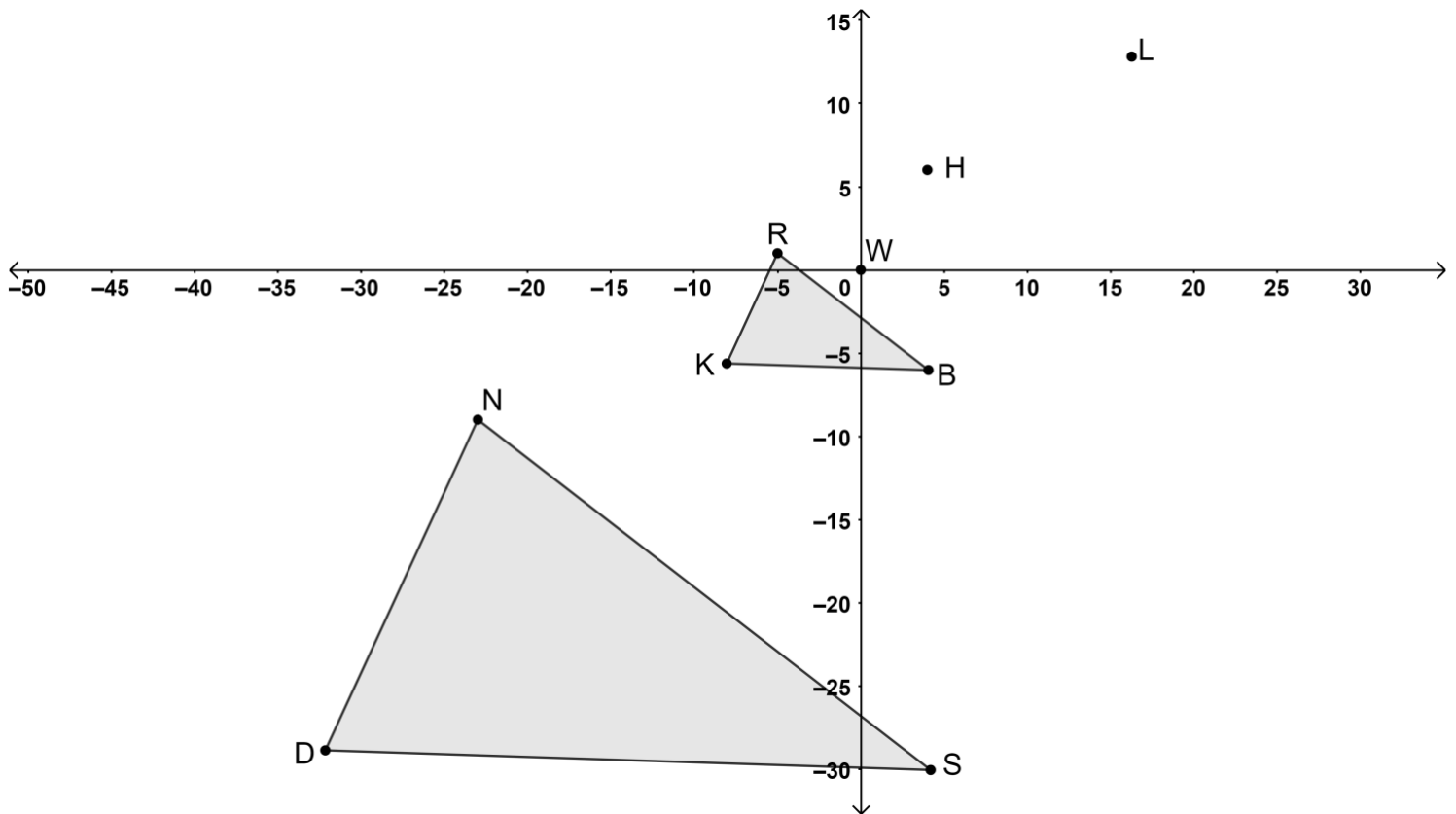


1. Triangles DNS and KRB are shown on the coordinate grid.



Complete the statement.

Triangle

- ☐ DNS
☐ KRB

is the result of dilating triangle

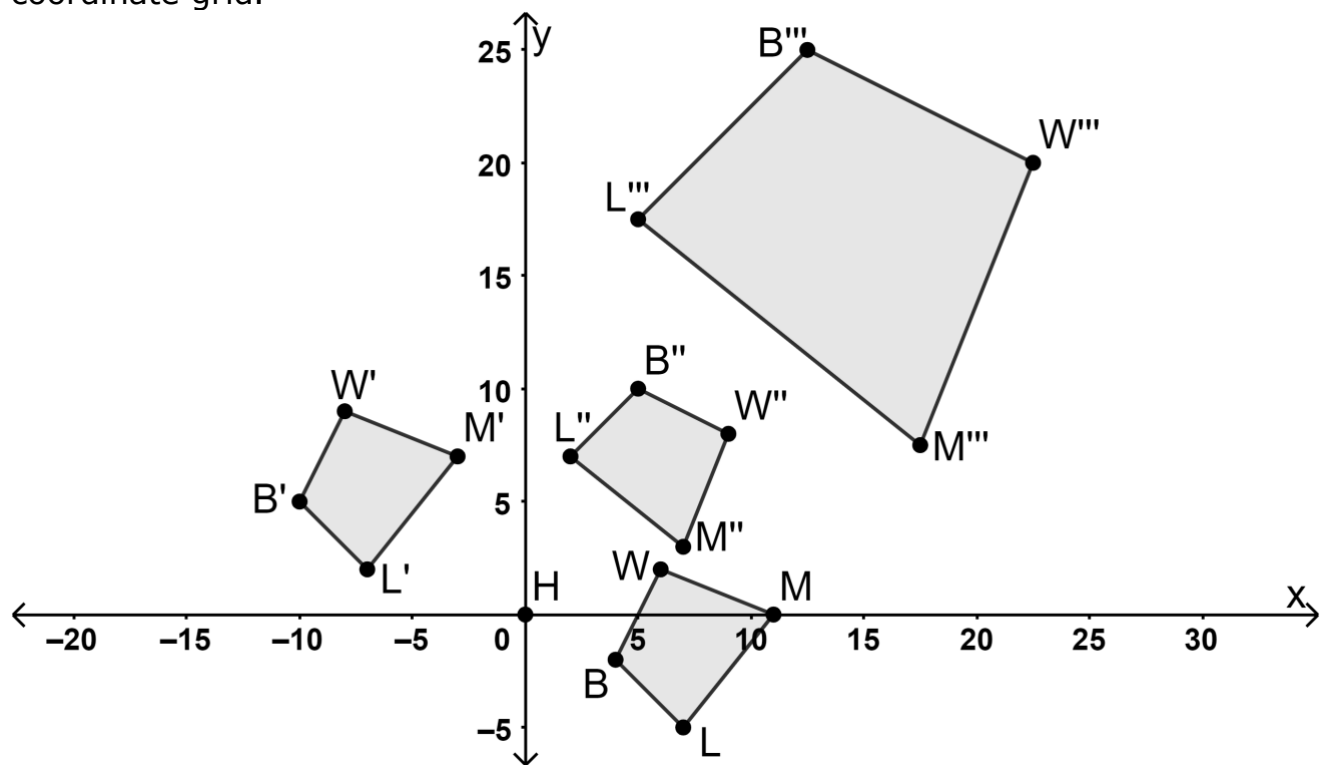
- ☐ DNS
☐ KRB

using a scale

factor of 3 centered at

- ☐ $H.$
☐ $L.$
☐ $W.$

2. Quadrilaterals $BLMW$, $B'L'M'W'$, $B''L''M''W''$, and $B'''L'''M'''W'''$ are shown on the coordinate grid.



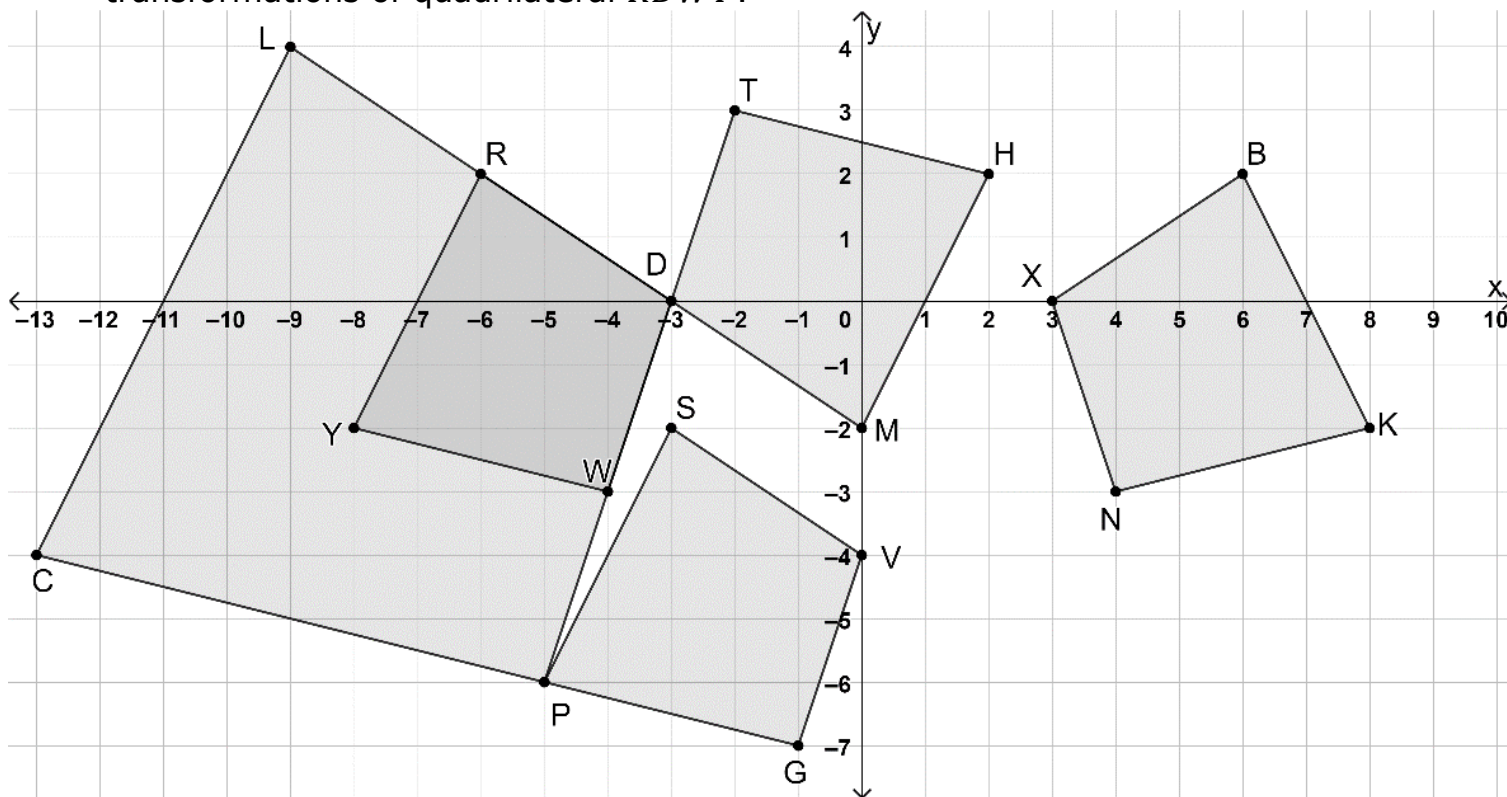
The coordinates of the preimage and each transformation are shown.

Preimage		Transformation 1		Transformation 2		Transformation 3	
B	$(4, -2)$	B'	$(-10, 5)$	B''	$(5, 10)$	B'''	$(12.5, 25)$
L	$(7, -5)$	L'	$(-7, 2)$	L''	$(2, 7)$	L'''	$(5, 17.5)$
M	$(11, 0)$	M'	$(-3, 7)$	M''	$(7, 3)$	M'''	$(17.5, 7.5)$
W	$(6, 2)$	W'	$(-8, 9)$	W''	$(9, 8)$	W'''	$(22.5, 20)$

Complete the table by describing each transformation and giving the algebraic description using coordinates.

	Written Description	Algebraic Description
Transformation 1	Translation left 14, up 7	$(x, y) \rightarrow (x - 14, y + 7)$
Transformation 2	Rotation 90° clockwise about the origin	$(x, y) \rightarrow (y, -x)$
Transformation 3	Dilation of 2.5 centered at the origin	$(x, y) \rightarrow (2.5x, 2.5y)$

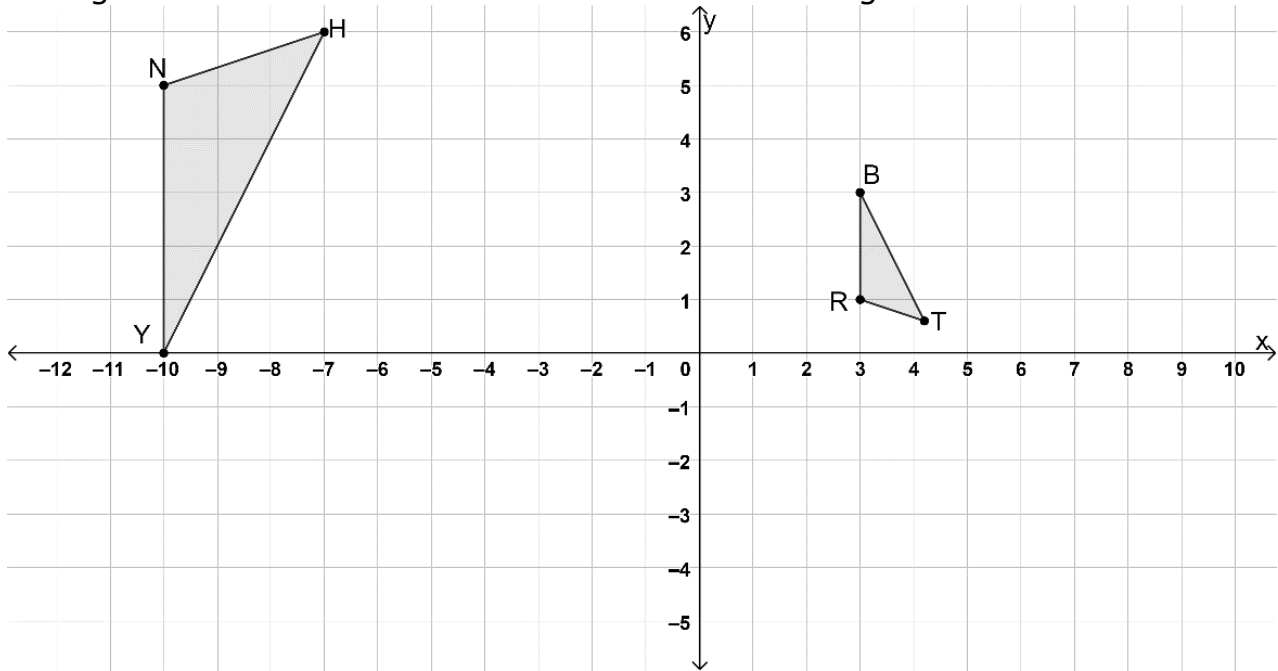
3. Quadrilaterals $RDWY$, $LDPC$, $TDMH$, $BXNK$, and $SVGP$ are shown on the coordinate grid, where quadrilaterals $LDPC$, $TDMH$, $BXNK$, and $SVGP$ are all transformations of quadrilateral $RDWY$.



Match each description with the quadrilateral that is the result of the transformation of quadrilateral $RDWY$. If none of the quadrilaterals match, then choose "none".

Transformation of quadrilateral $RDWY$	$LDPC$	$TDMH$	$BXNK$	$SVGP$	none
translation down 2	☐	☐	☐	☐	☒
translation right 6	☐	☐	☐	☐	☒
translation right 3 down 4	☐	☐	☐	☒	☐
dilation by a factor of 2 centered at point D	☒	☐	☐	☐	☐
dilation by a factor of $\frac{1}{2}$ centered at the origin	☐	☐	☐	☐	☒
reflection across y -axis	☐	☐	☒	☐	☐
reflection across $x = -3$	☐	☐	☐	☐	☒
rotation 180° about point D	☐	☒	☐	☐	☐
rotation 180° about the origin	☐	☐	☐	☐	☒

4. Triangles NHY and RBT are shown on the coordinate grid.



- a. Select all of the sequences of transformations that will result in the triangles being mapped onto each other.

- ☐ $\triangle BRT$ maps to $\triangle YNH$ using $(x, y) \rightarrow (-y, x)$ then $(x, y) \rightarrow (x - 3, y - 4)$ then $(x, y) \rightarrow (2x, 2y)$
- ☒ $\triangle BRT$ maps to $\triangle YNH$ using $(x, y) \rightarrow (x - 7, y - 3)$ then $(x, y) \rightarrow (x, -y)$ then $(x, y) \rightarrow (2.5x, 2.5y)$
- ☐ $\triangle BRT$ maps to $\triangle YNH$ using $(x, y) \rightarrow (-y, -x)$ then $(x, y) \rightarrow (5x, 5y)$ then $(x, y) \rightarrow (x, -y)$
- ☐ $\triangle YNH$ maps to $\triangle BRT$ using $(x, y) \rightarrow (x + 6, y)$ then $(x, y) \rightarrow (x, -y)$ then $(x, y) \rightarrow (0.5x, 0.5y)$
- ☒ $\triangle YNH$ maps to $\triangle BRT$ using $(x, y) \rightarrow (0.4x, 0.4y)$ then $(x, y) \rightarrow (x, -y)$ then $(x, y) \rightarrow x + 7, y + 3)$

- b. Using one of the mappings selected in part a, justify that triangles BRT and YNH are similar.

The mapping of triangle BRT to triangle YNH includes a dilation, so therefore the triangles are similar.

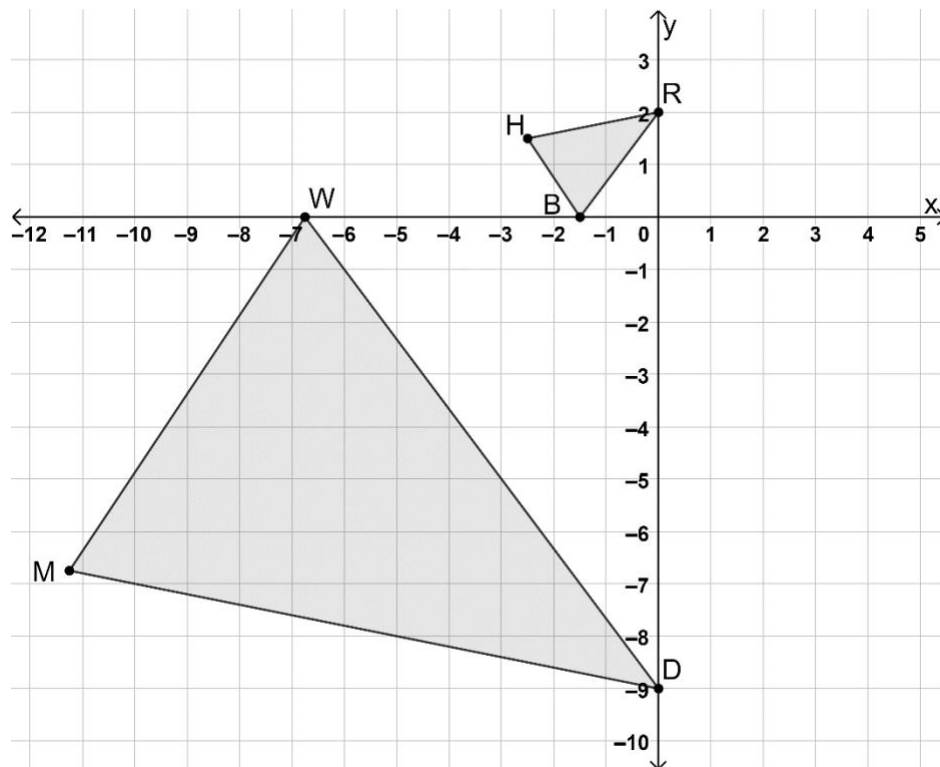
OR

The mapping of triangle YNH to triangle BRT includes a dilation, so therefore the triangles are similar.

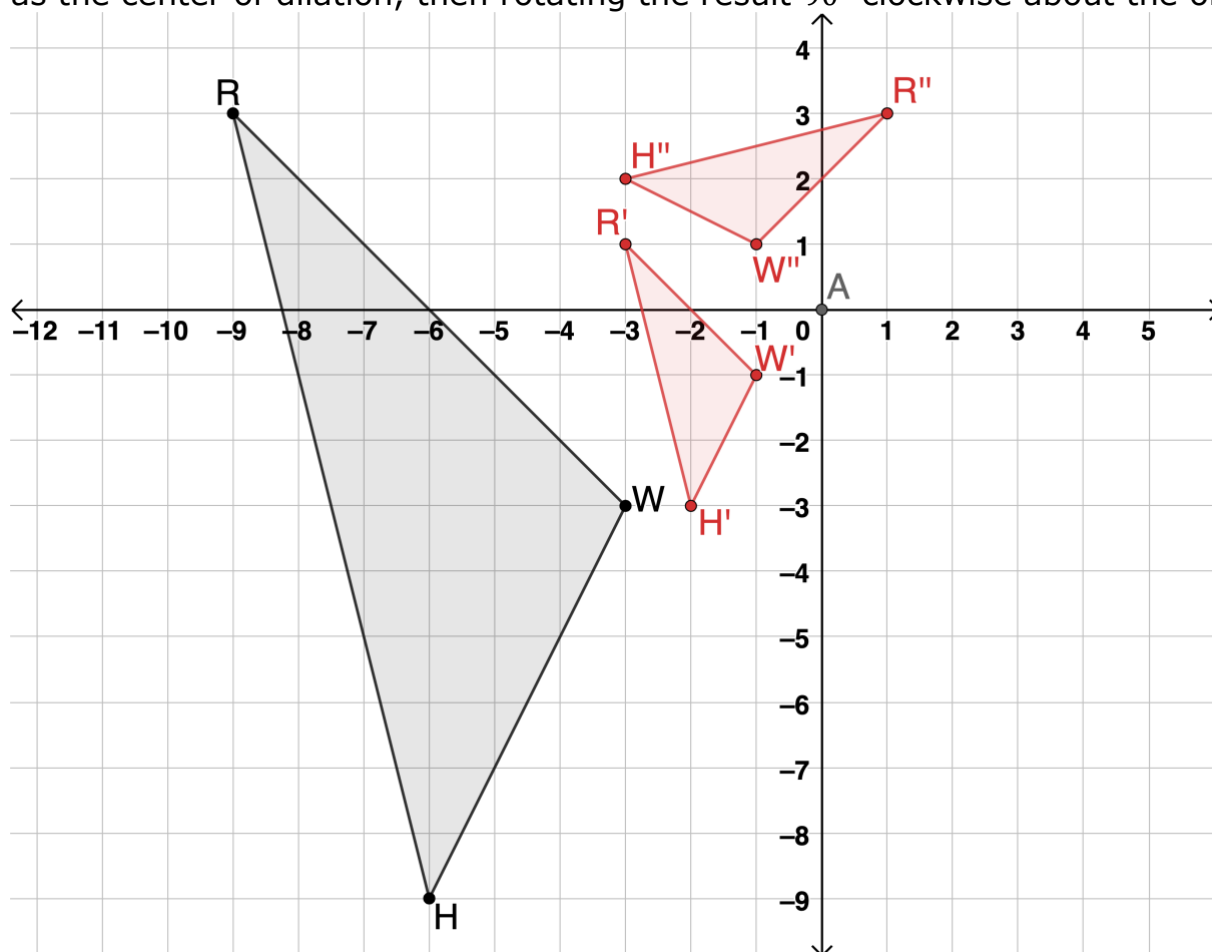
5. Triangles RHB and DMW are shown on the coordinate grid.

Write a sequence of transformations that will map $\triangle RHB$ onto $\triangle DMW$.

Reflect $\triangle RHB$ across the x -axis, then dilate using a scale of 4.5 centered at the origin.

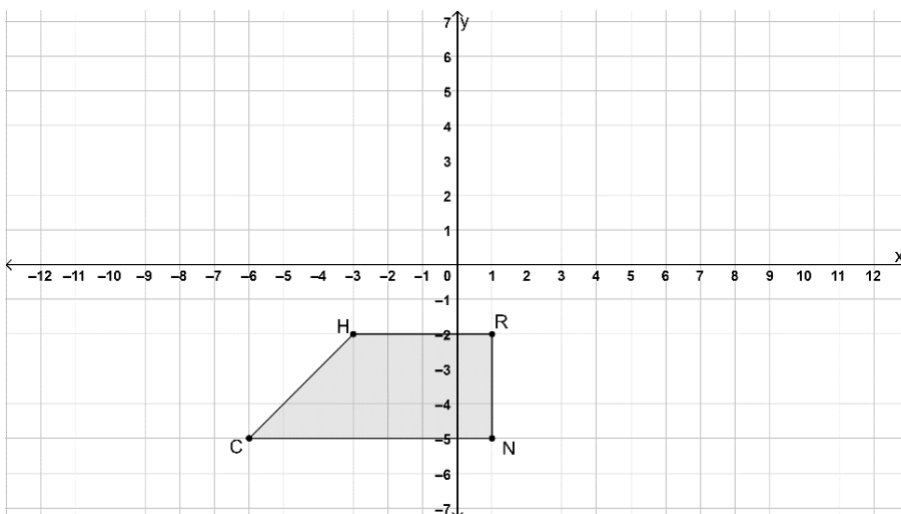


6. Triangle RWH is shown on the coordinate grid. On the same coordinate plane, draw $\triangle R''W''H''$, which is the result of dilating $\triangle RWH$ using a scale factor of $\frac{1}{3}$ with the origin as the center of dilation, then rotating the result 90° clockwise about the origin.



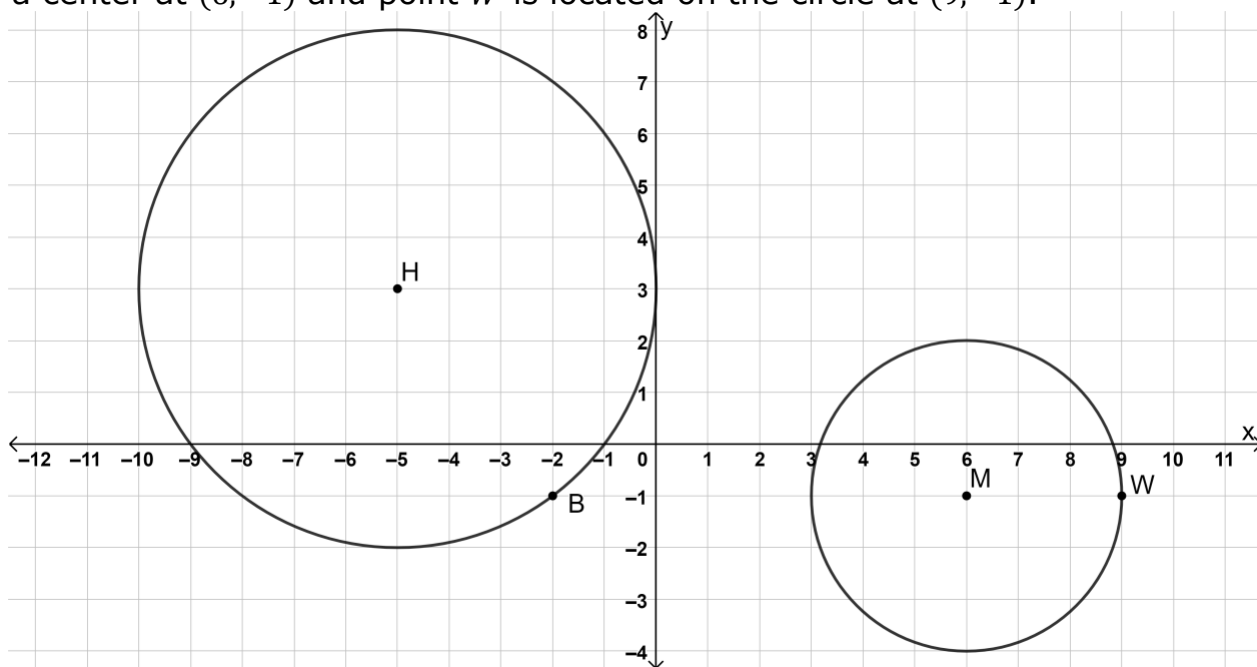
7. Quadrilateral $HRNC$ is shown on the coordinate grid.

Match each sequence of transformations of quadrilateral $HRNC$ with the characteristic(s) of $H''R''N''C''$, the result of the sequence of transformations.



Sequence of Transformations	Preserves Distance	Preserves Angle Measure	$H''R''N''C''$ is similar but not congruent to $HRNC$
$(x, y) \rightarrow (-y, x)$ then $(x, y) \rightarrow (3.5x, 3.5y)$	☐	☐	☐
$(x, y) \rightarrow \left(\frac{2}{3}x, \frac{2}{3}y\right)$ then $(x, y) \rightarrow (x, -y)$	☐	☐	☐
$(x, y) \rightarrow (x - 1, y + 5)$ then $(x, y) \rightarrow (x, y)$	☐	☐	☐
$(x, y) \rightarrow (-x, -y)$ then $(x, y) \rightarrow (x + 6, y - 2)$	☐	☐	☐

8. Circle H has a center at $(-5, 3)$ and point B is located on the circle at $(-2, -1)$. Circle M has a center at $(6, -1)$ and point W is located on the circle at $(9, -1)$.



Lakesha performs two transformations on circle H to show that circle H is similar to circle M .

- a. Complete the statements that describe Lakesha's transformations.

Lakesha first dilates circle H with a center of dilation at point H using a scale

factor of

- ☐ $\frac{1}{2}$.
- ☐ $\frac{2}{5}$.
- ☒ $\frac{3}{5}$.
- ☐ $\frac{5}{3}$.
- ☐ $\frac{5}{2}$.
- ☐ 2.

She then translates the circle

- ☐ right 11.
- ☐ right 6 down 3.
- ☒ right 11 down 4.
- ☐ right 8.5 down 2.5.

Lakesha then wrote this statement.

Circle H is not similar to circle M .

b. Complete the proof by contradiction.

Assume that circle H

- is
- is not

 similar to circle M . By definition of similarity,

two figures are similar if and only if one figure can be obtained from the other by

a

- dilation,
- rotation,
- reflection,
- translation,

 or a sequence of transformations that includes a

- dilation.
- rotation.
- reflection.
- translation.

Because one of the transformations is a

- dilation,
- rotation,
- reflection,
- translation,

the circles are mapped using

- rigid motions.
- similarity transformations.

 Therefore, circle H

is similar to circle M , which contradicts the given statement.

c. Complete the statement.

All circles

- can
- cannot

 be mapped to each other using

- rigid motions.
- similarity transformations